

Taal

The agent that sells what the forecast says you'll throw away.

TEAM

Nandish · lead

Rinu · co-builder

PROBLEM

- 1 Forecasts and offers live in different tools: one dark store writes stock off while the next runs out.
- 2 Customers get blanket markdowns, not offers matched to what they buy.
- 3 Nothing plans to the online sell-by cut-off: 30% of shelf life or 45 days at delivery.¹

¹ FSSAI advisory to e-commerce FBOs, 3 Dec 2024, file RCD-13/1/2024-Regulatory-FSSAI(E-13150): minimum 30% shelf life or 45 days at delivery. Either-limit reading is Taal's configurable default.

BRIEF ABOUT THE IDEA

One decision loop turns a forecast into a sale, before the write-off happens.

- 1 Capture** a pallet photo
- 2 Sense** nightly gaps, in rupees
- 3 Plan** a play; the estimator owns the rupees
- 4 Approve** with a holdout; feed it back to the forecast
- 5 Engage** consenting customers, in their language
- 6 Measure** treated vs holdout, 95% CI

THEME Taal forecasts stock per SKU and node nightly, with festival effects, and drafts human-approved offers.

ONE EXAMPLE · SEEDED DEMO TENANT

6 days

until Masala Chips at dark store DS-07 can no longer be sold online

368
units

₹9,200
at stake

Recorded Gemini run, 10 Oct: one schema rejection, then all eight guardrails pass. Approve: 323 treated, 29 holdout.

DIFFERENTIATION

A holdout on every play. The play written back into the forecast.

	ONLINE SELL-BY CUT-OFF	HOLDOUT PER PROMO	PROMO → FORECAST WRITE-BACK	INDIA / Q-COMMERCE
Taal	Yes	Yes	Yes	Built for it; seeded tenant
Blue Yonder	Unclear	Unclear	Unclear	Partial
RELEX	Partial	Unclear	Partial	Partial
Wasteless	No	Unclear	Unclear	Unclear
Afresh	No	Unclear	Unclear	No
Flashfood	No	No	No	No

Unclear = no public evidence found in vendor pages read on 2026-10-09; it does not mean the feature is absent. Partial and No are our reading of those pages. Afresh and Flashfood are different product categories.

LIST OF FEATURES

Six capabilities, gated by rules a model never touches.



Vision intake

Photograph a pallet; dates and counts come back with confidence. Low confidence goes to a person.



Nightly Sense

Per-SKU, per-node forecasts. Six gap types, each with rupees and a deadline.



Planner Agent

ADK LoopAgent, max 3 iterations, six deterministic tools; a Cost Governor picks the gaps.



Approve + assignment

A human approves. Holdout by hash; offers go to treated customers only.



Customer Agent

Grounded in stock, offers and consent. Orders through an in-process MCP tool.

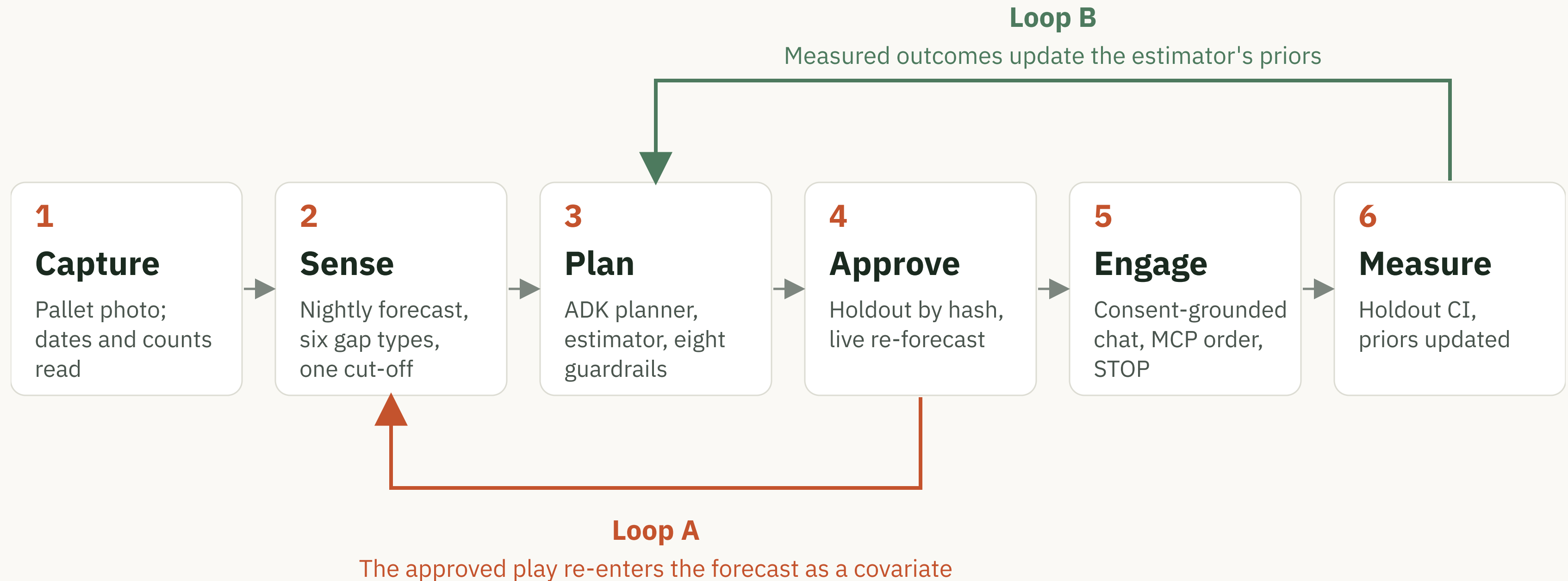


Measure

Treated vs holdout, 95% CI. Priors update from the counts.

PROCESS FLOW

Two closed loops, not an open-ended pipeline.



SCREENS

Guardrails reject; customers get Kannada plus English.

Recorded from Gemini · 28 Sept 2026

run run_chips_ds07_v1_e9971541

Replay at 4x

Guardrail attempts · 4 proposed, 3 rejected

Attempt 1

REJECTED

+25880ms

cite_or_drop

uncited numbers in rationale: 52

▼ Rejected rationale

This play proposes a bundle of "Masala Chips 200G" with "Coconut Water 1L" at a price of 58.50 INR, offered through an outlet channel. This approach is preferred over a markdown for an online_sellby_breach gap, aligning with policy preference for bundles and transfers. The play targets all available segments (totaling 334 customers after consent) and is expected to clear 22.98 units, generating an expected margin of 191.88 INR. This play passes all guardrails, including maintaining a net margin of 14.27% which is above the 8.00% category floor for snacks. The alternative transfer play was rejected due to a negative expected margin of -161.52 INR.

Attempt 2

REJECTED

+40063ms

cite_or_drop

uncited numbers in rationale: 52

► Rejected rationale

Attempt 3

REJECTED

+55016ms

cite_or_drop

uncited numbers in rationale: 52

► Rejected rationale

Attempt 4

PASSED ALL GUARDRAILS

+63144ms

▼ Accepted rationale

This play proposes a bundle of "Masala Chips 200G" with "Coconut Water 1L" offered through an outlet channel. This approach is preferred over a markdown for an online_sellby_breach gap, aligning with policy preference for bundles and transfers. This play passes all guardrails. The alternative transfer play was rejected.

Chat as Meena

LIVE

2482 ms

Customer Meena (DS-07, kn) ▼

treated on the chips play once approved

ನಮಸ್ಕಾರ, ಇಂದು ಯಾವುದಾದರೂ ಆಫರ್ ಇದೆಯೇ?

ನಮಸ್ಕಾರ! ಹೌದು, ಮಸಾಲಾ ಚಿಪ್ಸ್ 200G ಬಂಡಲ್ ಮೇಲೆ ಆಫರ್ ಇದೆ. ಇದರ ಬೆಲೆ ₹58.5. 2026-10-15 ರೊಳಗೆ ಇದನ್ನು ಬಳಸಿದರೆ ಉತ್ತಮ. ಈಗಲೇ ಆರ್ಡರ್ ಮಾಡಲು 'Add to cart' ಬಟನ್ ಒತ್ತಿ.

Add to cart

Not now

STOP

English: Hello! Yes, there is an offer on Masala Chips 200G bundle. Its price is ₹58.5. It is best to use it by 2026-10-15. To order now, press the 'Add to cart' button.

Do you have Cola Zero?

Type a message...

Send

STOP

Right: customer chat, live

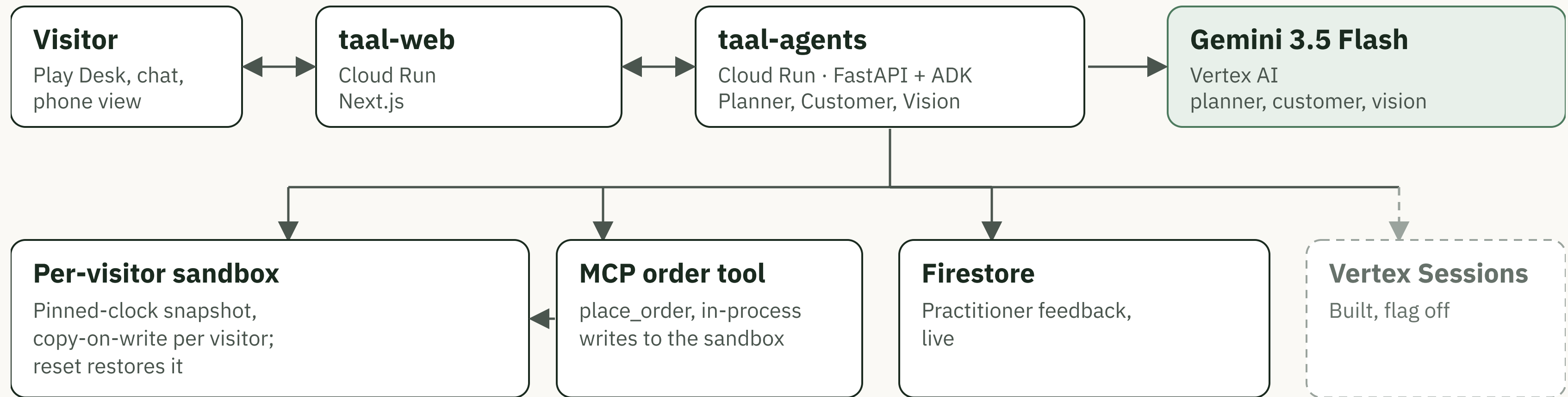
Kannada reply with an English gloss; the order goes through the MCP order tool.

Left: Play Desk trace, recorded run (28 Sep)

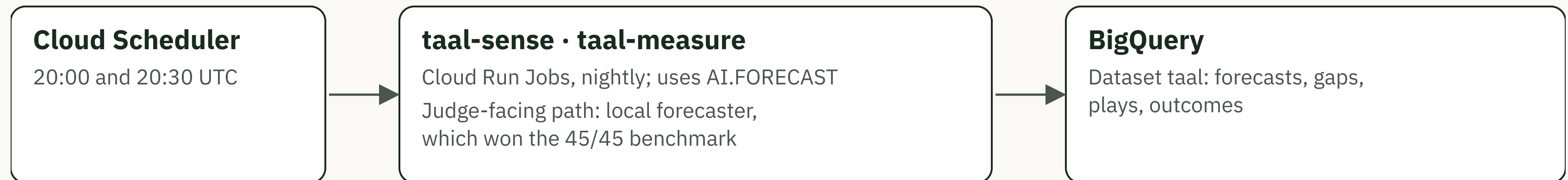
28 Sep run, before the tokenizer fix: three attempts rejected by cite-or-drop, the fourth passes. The 10 Oct run needed one schema retry.

ARCHITECTURE

A sandbox serves judges; nightly jobs run on BigQuery.



NIGHTLY PATH



TECHNOLOGIES USED

Google Cloud AI stack, benchmarked before use.

Gemini 3.5 Flash

On Vertex AI: vision intake, Planner Agent and Customer Agent.

Google ADK

LoopAgent (max 3), 6 deterministic tools and 8 guardrails around every play.

BigQuery

AI.FORECAST benchmarked and run nightly; AI.GENERATE_TABLE writes offer copy. Judges see the local forecaster.

Firestore

Practitioner feedback store, live in the deployed service.

Cloud Run

Services scale to zero; Cloud Run Jobs and Cloud Scheduler run Sense and Measure nightly.

Next.js · FastAPI · MCP

Play Desk, phone view and chat on Next.js; API on FastAPI; orders via an in-process MCP tool.

DEPLOY GitHub Actions runs verify, then deploys to Cloud Run on every merge to main.

ESTIMATED IMPLEMENTATION COST

Priced from measured usage; sold on rupees rescued.

PER 30 DAYS, MODELLED FROM MEASURED USAGE

₹486 / \$5.86

Real Gemini tokens, 752 BigQuery jobs and Cloud Run time, priced at public list rates. Not a bill.

Method: modeled_from_measured_usage.

Forward-looking profiles ESTIMATES

Profile	Scale	Monthly
Micro	1 shop, ≤500 SKUs	₹200–500
Mid	300 SKUs, 10 nodes	₹2,500–5,000
Large	20,000 SKUs, 500 nodes	₹1–2 lakh

BUYER AND PRICING

Buyer: the category or supply-chain head, who owns write-off.

Basis: a fee per node per month, or a percentage of rupees rescued vs the holdout.

BREAK-EVEN ILLUSTRATION

≈ ₹21,710 – ₹30 per node

Per month: projected margin ₹347,356 and run cost ₹486, each over 16 nodes. Synthetic, not a quote.

Doubtful reads go to a person; outcomes say SYNTHETIC.



SKU GUESS	BEST BEFORE	FACINGS	CONFIDENCE
SKU-WHEAT-ATTA-1KG	2026-11-05	4	sku 80% · date 70% · count 90%
SKU-POHA-500G	2026-10-25	4	sku 75% · date 65% · count 85%

Is this Poha 500G, best before 25 Oct 2026, 4 facings?

Yes

No

Confirm rows

Phone view

A read under 70% confidence is asked back before it reaches inventory.

Measured

Holdout is a randomly assigned control group -- customers who see no offer for this play at all, chosen the same way as everyone in "Treated" except for a coin flip. Every lift number below is the difference between what the treated group actually did and what this control group did, not a before/after comparison against the same customers. This is the only honest way to know a play caused the change, rather than a trend that would have happened anyway.

0.4 kg

of food diverted from waste, across 2 measured plays

1.0 kg

CO2e avoided (estimate; see the emissions factor caveat in docs/DATA_MODEL.md)

₹50

margin preserved that would otherwise have been written off

Run Measure

2 plays joined against orders: 2 measured, 0 unmeasured (below min treated n) · 12 Sept, 09:00 am

PLAY	TREATED	HOLDOUT	LIFT	WASTE AVOIDED	CEO NUMBER	DATA	LOOKER
SKU-MASALA-CHIPS-200G DS-07 · bundle min treated n 20 · measured 12 Sept, 09:00 am	Treated 316 customers, 1 responders, 2 units margin ₹9, discount ₹7	Holdout 36 customers, 0 responders, 0 units margin ₹0, discount ₹0	0.3% (CI -9.3%–1.8%)	₹50 0.4 kg	1.31 net margin recovered per ₹1 of discount vs holdout	• SYNTHETIC	Looker (n/a)
SKU-KAJU-KATLI-250G DS-03 · preorder min treated n 20 · measured 12 Sept, 09:00 am	Treated 295 customers, 0 responders, 0 units margin ₹0, discount ₹0	Holdout 28 customers, 0 responders, 0 units margin ₹0, discount ₹0	0.0% (CI -12.1%–1.3%)	₹0 0.0 kg	unmeasured	• SYNTHETIC	Looker (n/a)

Outcomes: treated vs holdout, SYNTHETIC badge

Screenshot from the 28 Sep build (316 / 36). The current seeded example is 323 treated, 29 holdout; the lift interval crosses zero and is reported as such.

PERFORMANCE REPORT

Measured, dated, and traced to a raw file.

PLANNER, LIVE10 Oct 5/5 proposed, 0 fallbacks; 22–28 s each eval/raw/planner_real_traces_2026-10-10/summary.json	CUSTOMER AGENT28 Sep 4.56 s p50 p95 6.48 s, 0.48 s over the 6 s budget; 50/50 returned 200 eval/raw/customer_latency_2026-09-28/analysis.json	APPROVE + RE-FORECAST21 and 28 Sep 8.3–10.6 s two runs; write-off ₹9,194 → ₹8,068 eval/raw/sweep_vertex_2026-09-21.txt · eval/raw/post_deploy_check_2026-09-28/summary.json	AGENT SIMULATION21 Sep 190/190 live Vertex; stock, holdout arm, stacking, STOP, hallucination eval/raw/agent_simulation_2026-09-21.json
VISION INTAKE21 Sep 30/30 correct dates; synthetic photos only eval/raw/vision_synthetic_2026-09-21.json	BACKTEST, SYNTHETIC20 Sep 0.113 WAPE MAPE 0.133 over 63 tier-origin rows eval/raw/backtest_rows_2026-09-20.jsonl · eval/raw/external_backtest_2026-10-09.json	PUBLIC NON-GROCERY DATASET BACKTESTUCI Online Retail II 0.83 WAPE Method and caveats in eval/external_backtest.md eval/raw/external_backtest_2026-10-09.json	
LIVE RE-PLAN, DEPLOYED10 Oct 30.3 / 27.5 / 29.6 s 3 live runs, 0 fallbacks, 1 loop iteration each; 90 s deadline eval/raw/live_replan_timing_2026-10-10.json · eval/raw/live_replan_timing_2026-10-09.json	47.7 / 47.7 / 30.8 s before the fix, 9 Oct: 2 of 3 hit a 45 s deadline and fell back to rules	LIVE EVAL ON VERTEX2026-10-09 15/15 rationale judge; same-family judge, weak evidence eval/raw/live_eval_2026-10-09/summary.json 38/46 ADK response match 6/46 exact trajectory; retry-order variance	

What seven practitioners told us.

3 of 7

write off stock weekly or daily

3 of 6

online sellers say the sell-by rule limits what they sell

0 of 7

measure a promotion against a control group

4 of 7

want a one-week pilot

"Blanket markdowns erode margin on products that may still sell normally."

Category / merchandising, quick-commerce

"Item-store recommendations with incremental-margin estimates."

Category / merchandising, quick-commerce

7 usable responses (10 collected, 3 excluded as flagged in config/feedback_exclusions.json). Anonymous by consent; contacts never published. Counts, not percentages.

Three people, three screens, one approved play.

Planner

/desk

- 1** Open the Play Desk and pick a gap.
- 2** Check the guardrails and counterfactuals.
- 3** Click Approve. The forecast moves.

Node manager

/phone

- 1** Photograph a pallet, or tap a sample.
- 2** Confirm the rows read below 0.7 confidence.
- 3** Approve the play from the gap card.

Customer

/chat

- 1** Open the chat. It starts with "Any offers today?".
- 2** Ask about a product. You see a stock band, not a count.
- 3** Send STOP to withdraw consent.

Badges. LIVE: a real call this visit. REPLAY: stored output replayed. SYNTHETIC: fabricated demo data. "Recorded from Gemini": a dated real model run.

Projected margin hinges on one assumption.

TWO LOTS, 10 OCT Tea (the 60-second beat) +₹4,493 103 units, ₹35,020 at stake. Play net –₹27,522 against –₹32,015 for a blanket 20% markdown. Chips (recorded Gemini run) -₹279 368 units, ₹9,200 at stake. Play net –₹8,468 against –₹8,189. The Desk shows the loss.	RESPONSE PRIOR SENSITIVITY · PROJECTED <table><tr><th>PRIOR</th><th>MARGIN</th><th>WASTE AVOIDED</th></tr><tr><td>2%</td><td>₹1.41 lakh</td><td>₹3.26 lakh</td></tr><tr><td>5% default</td><td>₹3.47 lakh</td><td>₹7.06 lakh</td></tr><tr><td>10%</td><td>₹6.19 lakh</td><td>₹11.18 lakh</td></tr></table> Seeded tenant, 28-day horizon, 550 grocery gaps; 421, 424 and 423 plays planned.	PRIOR	MARGIN	WASTE AVOIDED	2%	₹1.41 lakh	₹3.26 lakh	5% default	₹3.47 lakh	₹7.06 lakh	10%	₹6.19 lakh	₹11.18 lakh	MEASURED VS PROJECTED Measured on real customers: none yet. Synthetic holdout: 323 treated, 29 held back; lift +1.5 points, 95% CI –12.4 to +5.6, crosses zero. Projected at 5%: ₹347,356 margin, ₹706,183 waste avoided.
PRIOR	MARGIN	WASTE AVOIDED												
2%	₹1.41 lakh	₹3.26 lakh												
5% default	₹3.47 lakh	₹7.06 lakh												
10%	₹6.19 lakh	₹11.18 lakh												

Practitioners: 7 usable responses; 3 of 7 write off stock weekly or daily.

What it takes to run past the competition.

Production path

Least-privilege accounts replace the deploy owner.

CI uses a stored key. Move to workload identity.

Secrets sit in a managed store.

Model IDs pinned with retirement dates: gemini-3.5-flash live, Flash-Lite fallback; the previous model retires 20 Oct 2026.

Scalability

Measured: Sense 3.8 s; planner re-plan 27 to 30 s live; chat 4.5 to 6.9 s.

Estimate: 20,000 SKUs, 500 nodes.

ESTIMATE

Breaks first: the planner queue.

Feasibility

Buyer: category or supply-chain head.
Fee per node or rupee share.

Pilot: 50 to 60 testers, one week, not yet run.

Run cost: ₹486 a month, measured usage, not a bill.

Estimates: ₹200 to 500 (Micro), ₹2,500 to 5,000 (Mid), ₹1 to 2 lakh (Large). ESTIMATE

Next: a real pilot, real photos, more tenants.

1

A pre-registered pilot

One week, 50–60 consented testers, 40–50% holdout.
Metric pre-registered in docs/pilot.md.

DESIGNED

2

Vision on real photos

Harness built. Real camera photos are the missing input; 30/30 is on synthetic ones.

HARNESS BUILT

3

Multi-tenant BigQuery

Tenant id on every table; the BigQuery path is built, flag off until a second tenant.

BUILT, FLAG OFF

LINKS

Taal: the agent that sells what the forecast says you'll throw away

GITHUB REPOSITORY



<https://github.com/Nandish3010/probable-fortnight>

LIVE APP



<https://taal-web-2obkp776ca-el.a.run.app>

DEMO VIDEO

QR added with the video link

link added at submission